VIJAY BOMMIREDDY

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EDUCATION:

Indiana University | MS in Data science | Indianapolis, IN — Jan 2023-Dec 2024

Coursework : *Intro to Informatics, Statistics, Data Analytics, Machine Learning, deep learning and Cloud Computing.* — GPA: 3.97/4.0

WORK EXPERIENCE:

Lifeline of Ohio| Columbus, OH — Feb 2025- *present*

Business Intelligence Analyst

- Launched 60+ Power BI and Tableau dashboards by establishing a structured requirements-gathering process, cutting turnaround time by 40%, scaling delivery capacity by 35% and automating cross-functional workflows via **n8n** to eliminate manual operational overhead.
- Incorporated **SiEBERT** model on external organizational surveys, improving strategic resource allocation accuracy by 22% and enabling leadership to resolve critical engagement gaps 3x faster.
- Streamlined an end-to-end MLOps pipeline integrating **DARTS** time-series forecasting with MLflow and Apache Airflow, automating drift detection and model retraining cycles that eliminated 60% of manual intervention, saving 8+ hours weekly.
- Established a **Temporal Fusion Transformer** (TFT) for DCD time-to-death prediction, reducing false viability assessments by 30%, saving clinical teams significant evaluation time per case, and directly increasing viable organ donation rates.
- Engineered a quantized multi-modal LLM chatbot using **Agentic RAG** and REST-based MCP server integrations, operated on CPU saving 60% infrastructure costs, cutting analyst query resolution time by 50% and eliminating third-party BI tool dependency entirely.

Clear Object | Fishers, IN — Jan 2024 - May 2024

Data scientist Intern

- Collaborated with 6+ cross-functional teams (data engineers and cloud architects) to develop and fine-tune **YOLOv8** on a custom manufacturing image dataset for tracking cart movements and mandrel positioning.
- Integrated a **CI/CD** pipeline for data collection and model training using Vertex AI, lowered deployment time from 5 days to 3 days.
- Converted a segmentation model to ONNX format for optimized deployment on edge IoT devices using Docker, enabling real-time monitoring and reducing latency by 50% (from 200 ms to 100 ms).
- Implemented anomaly detection alerts, reducing maintenance response time and saving **$85,000** in emergency downtime costs.

Innovation, Incubation & Entrepreneurship Development Centre - NIT Srinagar | India — Dec 2021- Dec 2022

Data scientist

- Designed and integrated CMR data exchange and processing **APIs** to improve data pipeline efficiency, handling over 10 million rows and 200+ columns of real-time data enabling faster business analytics and building dashboards in Tableau.
- Programmed a macro using VBA to transform forecasted financial non-tabular data into tabular format, saving over 5 hours weekly.
- Orchestrated significant statistical analysis, increasing model reliability and validity via **hypothesis testing** and performance evaluation.
- Developed and Implemented **Ensemble** machine learning models for Research Publications, improving prediction accuracy by 15% and enhancing model performance with an optimized AUC-ROC curve for better classification results.

ACADEMIC PROJECTS:

Equipment Health Monitoring System [GitHub] [App]

- The failure classifier model, utilizing **data wrangling** and manipulation, enabled preventive maintenance, cutting costs by 35%.
- Applied SMOTENC for handling unbalanced classes, applied **PCA** for dimensionality reduction, and deployed the model using Streamlit, resulting in a 20% increase in user engagement**.**

Customer Segmentation[GitHub]

- Revised customer insights and accuracy by 15% with low **Type I/II errors**, leveraging feature engineering and high-quality data integration.
- Achieved 20% revenue growth by optimizing targeting strategies using KMeans++ clustering, ensuring high accuracy with improved silhouette scores for better segmentation and predictions.

Time series anomaly detection [GitHub]

- Explored API for statistical models and predictive models, detected real-time anomalies with 95% accuracy in noisy and mislabeled data.
- Automated resource management and computation workflows using **ARIMA** and Prophet models, reducing processing time by 40% while meeting precision, recall, and mean absolute error (MAE) standards for accurate time series forecasting.

SKILLS:

Programming : Python (OOPs, NumPy, Pandas, Scikit-learn), R, SQL and Spark.

Machine learning : Regression, Naive Bayes, Decision trees, SVM, Random Forest, Clustering, XGBoost and Neural Network.

Databases &Cloud : AWS(Elasticsearch, EC2, S3, Lambda), GCP(Vertex AI), Big Query, Docker, Snowflake, MySQL and SQL Server.

Tools : Tableau, Looker, Power BI, Microsoft Excel(Advance), Jira, SAS, Git, and SPSS.